

A Comparative Analysis of Machine Learning Architectures for siRNA Efficacy Prediction

A B.Tech Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology

by

Animesh kumar
(22CS30009)

Under the supervision of
Prof. Pralay Mitra



Department of Computer Science and Engineering

Indian Institute of Technology, Kharagpur

Kharagpur, India

October 30, 2025

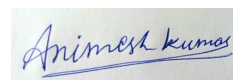
DECLARATION

I certify that

- (a) The work contained in this report has been carried out by us under the guidance of our supervisors.
- (b) The work has not been submitted to any other institute or university for the award of any degree or diploma.
- (c) We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and listing their details in the references. In addition, we have obtained permission from copyright owners whenever necessary.

Date: October 30, 2025

Place: Kharagpur



(Animesh Kumar)

Roll No. 22CS30009

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR - 721302, INDIA



CERTIFICATE

This is to certify that the project report entitled “**A Comparative Analysis of Machine Learning Models for siRNA Efficacy Prediction**” submitted by **Animesh Kumar** (Roll No. 22CS30009) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

Date: October 30, 2025
Place: Kharagpur

Professor Pralay Mitra
Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur
Kharagpur - 721302, India

Acknowledgements

I wish to extend my deepest appreciation to my project supervisor, Prof. Pralay Mitra. His mentorship was pivotal throughout the entire lifecycle of this research. His profound expertise in machine learning and insightful critiques were essential in navigating the complexities of comparing diverse model architectures for siRNA efficacy prediction. His guidance not only helped us refine our experimental methodology but also encouraged us to think critically about the trade-offs between model performance and complexity. We are truly grateful for his time and dedication to this project.

Abstract

The computational prediction of small interfering RNA (siRNA) efficacy is a critical task for advancing therapeutic gene silencing. While state-of-the-art approaches have utilized complex Graph Neural Networks (GNNs), this project investigates the performance of more established and interpretable machine learning architectures on a comprehensive, biologically informed feature set.

Leveraging the extensive feature engineering pipeline from the *siRNADiscovery* paper, we implement and conduct a rigorous comparative analysis of four distinct model families. We evaluate three powerful ensemble models on a flattened feature table: XGBoost, LightGBM, and Random Forest. Additionally, we introduce a hybrid **Spektral (GNN)** approach, which generates graph Spektral embeddings to encode the underlying data topology as features for an XGBoost model.

Our experiments, conducted across standardized data splits, reveal that the hybrid Spektral (GNN) model achieves the highest predictive accuracy among our implementations, with a Pearson Correlation Coefficient (PCC) of 0.729. This demonstrates the value of incorporating graph structural information. However, we find that standard gradient boosting models, particularly LightGBM (PCC of 0.718), offer highly competitive performance with significantly less implementation complexity. This work provides a clear, data-driven analysis of the trade-offs between feature representation, model architecture, and predictive power, highlighting the effectiveness of both ensemble methods and hybrid Spektral approaches for this important bioinformatics task.

Contents

Contents	5
List of Figures	7
List of Tables	8
1 Introduction and Motivation	9
1.1 The Big Picture: RNA Interference as a Therapeutic Tool	9
1.2 The Central Challenge: The Complexity of Predicting siRNA Efficacy	10
1.3 Research Question and Our Contribution	10
1.4 Report Structure	11
2 Literature Review	12
2.1 First-Generation Models: Empirical Rules	12
2.2 Advanced Architectures for Efficacy Prediction	13
2.2.1 Convolutional Neural Networks (CNNs)	13
2.2.2 Graph Neural Networks (GNNs)	13
2.2.3 High-Performance Ensemble Methods	13
2.3 Identifying the Research Gap	14
3 Methodology	15
3.1 Dataset and Experimental Setup	15
3.1.1 Dataset Provenance and Characteristics	15
3.1.2 Data Preprocessing and Splitting Strategy	15
3.2 Feature Engineering Pipeline	16
3.2.1 Non-Empirical Rule-Based Features	17
3.2.2 Empirical Rule-Based Features	18
3.3 Model Architectures and Implementation	19
3.3.1 Data Structuring for Different Models	19
3.3.2 Model Training Details	20
3.4 Evaluation Metrics	21

4	Results and Analysis	22
4.1	Experimental Setup and Evaluation Metrics	22
4.2	Comparative Performance Analysis	22
4.3	Discussion of Results	24
4.3.1	Overall Performance Landscape	24
4.3.2	Analysis by Metric	25
5	Conclusion and Future Work	26
5.1	Conclusion	26
5.2	Future Work	26
5.2.1	Advanced Hyperparameter Optimization	27
5.2.2	Deeper Investigation into Graph-Based Models	27
5.2.3	Feature Selection and Hybrid Models	27

List of Figures

4.1	Comparison of Pearson Correlation Coefficient (PCC) Across All Models.	23
4.2	Comparison of Spearman Rank Correlation Coefficient (SPCC) Across All Models.	23
4.3	Comparison of Mean Squared Error (MSE) Across All Models. . . .	24
4.4	Comparison of Area Under the ROC Curve (AUC) Across All Models.	24

List of Tables

4.1	Overall Performance of All Models (Averaged Across 3 Splits). The best result among our implemented models is shown in bold.	22
-----	--	----

Chapter 1

Introduction and Motivation

1.1 The Big Picture: RNA Interference as a Therapeutic Tool

RNA interference (RNAi) is a natural biological process for gene silencing that holds immense promise for modern medicine. It utilizes small interfering RNAs (siRNAs)—short, double-stranded RNA molecules, typically 19-23 nucleotides in length. These molecules act as guides for a cellular machine known as the RNA-Induced Silencing Complex (RISC). Once an siRNA is loaded into RISC, it directs the complex to its complementary sequence on a target messenger RNA (mRNA). The catalytic core of RISC, a protein called Argonaute-2 (AGO2), then cleaves the mRNA, leading to its degradation and effectively preventing the synthesis of the corresponding protein [1].

The ability to selectively "turn off" any gene of interest makes RNAi a profoundly powerful tool. It holds immense therapeutic potential for a vast spectrum of human diseases, including viral infections, genetic disorders, and various forms of cancer, by targeting the specific genes responsible for the pathology.

1.2 The Central Challenge: The Complexity of Predicting siRNA Efficacy

While the therapeutic promise of RNAi is vast, its practical application is bottlenecked by a significant design challenge. For any given mRNA target, thousands of potential siRNA candidates can be computationally designed. However, experimental validation reveals that their gene-silencing efficacy varies dramatically, with many candidates showing little to no activity. Manually screening this vast candidate pool is prohibitively expensive and time-consuming, making accurate computational prediction a vital necessity.

The difficulty in prediction stems from the multifaceted and non-linear nature of the gene silencing mechanism. The efficacy of an siRNA is not determined by its sequence alone, but by a complex interplay of numerous factors, including:

- **Positional Context:** The location of the siRNA target site on the mRNA strand is critical. The local secondary structure of the mRNA can either hide or expose the target sequence, significantly impacting the siRNA’s ability to bind [2].
- **Structural and Thermodynamic Properties:** The stability of the siRNA duplex itself, as well as the stability of the siRNA-mRNA complex, influences guide strand selection and the efficiency of RISC loading. Factors like base-pairing probabilities and Gibbs free energy play a crucial role [3].
- **Biophysical Interactions:** The entire process is mediated by proteins. The interaction affinity between the RNA strands and the key AGO2 protein is a fundamental biophysical constraint that has often been overlooked in earlier models [4].
- **Empirical Sequence Rules:** Through years of experimental work, researchers have identified numerous empirical rules. These include the impact of nucleotide frequencies (k-mers), G/C content percentages, and the specific nucleotides at key positions within the siRNA strand.

A successful predictive model must therefore be able to integrate these diverse and interacting features to accurately forecast the final silencing outcome.

1.3 Research Question and Our Contribution

The current state-of-the-art model, ‘siRNADiscovery’, addresses this complexity by using a Graph Neural Network (GNN) built upon a comprehensive feature set [5]. While theoretically powerful, this approach introduces significant complexity and yields predictions that are difficult to interpret. This leads to our central research question:

Given a rich and comprehensive set of biological features, is the complex relational learning of a GNN necessary for state-of-the-art siRNA efficacy prediction, or can alternative, more interpretable machine learning models achieve comparable performance?

Our contribution is a rigorous, empirical comparison of four distinct machine learning architectures. We faithfully reimplement the feature engineering pipeline from ‘siRNADiscovery’ and evaluate several powerful alternatives: XGBoost, LightGBM, Random Forest, and a novel hybrid Spektral (GNN) model. This study provides a data-driven analysis of the trade-off between model complexity, predictive accuracy, and interpretability.

1.4 Report Structure

The remainder of this report is organized as follows. Chapter 2 provides a critical review of the existing literature on siRNA prediction models. Chapter 3 details our experimental methodology. Chapter 4 presents and analyzes the performance of each model. Finally, Chapter 5 summarizes our findings and suggests directions for future work in this domain.

Chapter 2

Literature Review

The computational prediction of siRNA efficacy has evolved significantly over the past two decades. This evolution reflects both the growing availability of experimental data and the increasing sophistication of machine learning techniques. The models can be broadly categorized into three generations, each improving upon the last by incorporating more complex data and advanced algorithms.

2.1 First-Generation Models: Empirical Rules

The earliest predictive tools were based on empirical rules derived from the analysis of small, validated siRNA datasets. Researchers identified a set of sequence-based heuristics, often referred to as "design rules," that correlated with higher silencing activity. For example, the rules developed by Tuschl et al. provided guidelines on GC content, nucleotide preferences at specific positions (like the 5' and 3' ends), and the avoidance of certain sequence motifs [5].

Critical Evaluation: While these first-generation models were simple to implement and provided a valuable starting point, their predictive power was limited. Their reliance on small and often homogenous datasets meant they failed to capture the broader biological context and often did not generalize well to new targets or cell lines.

2.2 Advanced Architectures for Efficacy Prediction

2.2.1 Convolutional Neural Networks (CNNs)

To overcome the limitations of manual feature engineering, researchers began to apply deep learning. Han et al. [6] pioneered the use of a Convolutional Neural Network (CNN) for this task. CNNs are highly effective at automatically learning local patterns and sequence motifs from raw sequence data (e.g., one-hot encoded strings).

Critical Evaluation: While CNNs offered improved performance by automatically detecting predictive features, they primarily focus on local sequence context. They are less adept at modeling the global, long-range structural dependencies and the complex, non-linear interactions between the siRNA, the mRNA target site, and key proteins like AGO2.

2.2.2 Graph Neural Networks (GNNs)

The most sophisticated approach to date involves modeling the siRNA-mRNA interaction network as a graph. A notable early model, GNN4siRNA, introduced a heterogeneous graph with siRNA, mRNA, and interaction nodes [7]. However, this model relied solely on k-mers for sequence embedding and overlooked many of the critical biological features discussed in Chapter 1, such as thermodynamic stability, positional context, and RNA-protein interactions.

This limitation suggests that a state-of-the-art GNN should not only leverage the graph structure but also incorporate a comprehensive set of non-empirical and empirical features to create a more holistic model. Such a model, using an architecture like Heterogeneous GraphSAGE (HinSAGE) [8], is purpose-built to model complex data interactions and represents the current pinnacle of architectural complexity for this problem.

Critical Evaluation: The primary drawbacks of a full GNN approach are its high implementation complexity and its "black-box" nature, which makes it difficult to interpret which biological features are driving its predictions.

2.2.3 High-Performance Ensemble Methods

While deep learning models have gained prominence, ensemble methods remain the state-of-the-art for a wide range of structured (tabular) data problems. Models like **XGBoost**, **LightGBM**, and **Random Forest** are known for their high performance, robustness, and—in the case of tree-based models—their high degree of interpretability. These models operate on a flattened feature matrix and have not been systematically compared against feature-rich GNNs in the context of siRNA efficacy prediction.

2.3 Identifying the Research Gap

This review reveals a clear trajectory towards increasing model complexity, culminating in the feature-rich GNN concept. However, it also highlights that the feature sets used by these models have become progressively more sophisticated.

This leads to a critical, unaddressed gap in the literature: **a systematic comparison of advanced architectures on a level playing field**. It is unclear whether the superior performance of a GNN comes from its ability to learn from graph relationships or simply from the power of the extensive feature set it is trained on.

Our project is designed to fill this gap. We ask: **How do high-performance, interpretable ensemble models fare against a feature-rich GNN-based approach when both are given the exact same data?** Furthermore, we explore a novel hybrid approach—the Spektral (GNN) model—which attempts to capture the best of both worlds by encoding graph topology as features for an ensemble model, a strategy not previously applied in this specific context. By answering these questions, our work aims to provide a clear, data-driven analysis of the trade-offs between model architecture, performance, and interpretability.

Chapter 3

Methodology

This chapter details the systematic approach taken to evaluate the performance of different machine learning architectures for siRNA efficacy prediction. We first describe the dataset and preprocessing steps, followed by a comprehensive overview of the feature engineering pipeline adopted from siRNADiscovery [5]. Subsequently, we outline the specific implementation and training procedures for each of the models compared in this study. Finally, we define the metrics used for performance evaluation.

3.1 Dataset and Experimental Setup

3.1.1 Dataset Provenance and Characteristics

This study utilizes the **Dataset_HUVK**, a publicly available benchmark dataset compiled by Massimo et al. [7]. This dataset aggregates results from five foundational studies in the field of RNAi. It consists of 2,816 unique siRNA-mRNA pairs, each associated with an experimentally validated efficacy score. The efficacy is represented as a continuous value, typically ranging from 0 (no silencing) to 1 (complete silencing). The dataset includes 44 unique mRNA sequences and 2,816 unique siRNA sequences.

3.1.2 Data Preprocessing and Splitting Strategy

Prior to feature extraction, all RNA sequences containing uracil ('U') nucleotides were standardized to a DNA alphabet by converting 'U' to thymine ('T'). This allows for consistent processing with bioinformatics tools that primarily operate on DNA sequences. Any non-standard nucleotide characters (e.g., 'N') in the mRNA sequences were removed to ensure compatibility with feature extraction algorithms.

To ensure a robust and unbiased evaluation, we adopted the data-splitting methodology proposed by the 'siRNADiscovery' authors, which is designed to prevent data leakage. The dataset was randomly partitioned into a training set (70%), a validation (or development) set (15%), and a test set (15%). This process was repeated three

times using different random seeds to create three independent splits. All models were trained and evaluated on each of the three splits, and the final reported metrics are the average of these three runs. This approach mitigates the risk of a single, favorable data split skewing the results and provides a more reliable measure of each model’s generalization capability.

3.2 Feature Engineering Pipeline

Our methodology is founded on the faithful reimplementation of the comprehensive feature extraction pipeline from ‘siRNADiscovery’ [5]. This process generates a high-dimensional feature vector for each siRNA-mRNA interaction by combining features derived from the individual molecules and their predicted interplay. These features are categorized into two main types: non-empirical and empirical rule-based features.

3.2.1 Non-Empirical Rule-Based Features

These features are derived directly from the primary sequence and predicted structural properties of the RNA molecules.

[label=a)]

1. **Sequence Embedding:** Standard one-hot encoding was employed to convert the RNA sequences into a numerical format suitable for machine learning. Each nucleotide in the siRNA sequence (length 21) and the mRNA target region (length 38) was represented by a 4-dimensional binary vector: A = [1,0,0,0], C = [0,1,0,0], G = [0,0,1,0], and T = [0,0,0,1].
2. **Positional Embedding:** The position of the siRNA target site along the mRNA strand is a crucial factor in silencing efficacy. This spatial context was captured using a positional encoding matrix based on sine and cosine functions of varying frequencies, as described in the original paper.
3. **RNA Base-Pairing Probabilities:** RNA secondary structure significantly influences its function. Using the ViennaRNA package, we calculated base-pairing probability matrices for three scenarios: (1) the siRNA alone (self-fold), (2) the mRNA target site alone (self-fold), and (3) the siRNA-mRNA duplex (co-fold). Due to the high dimensionality and sparsity of these matrices, Truncated Singular Value Decomposition (SVD) was applied to reduce them to fixed-size feature vectors.
4. **RNA-Protein Interaction Probabilities:** The interaction between RNA strands and the Argonaute-2 (AGO2) protein is fundamental to the RNAi process. We utilized the RPISeq tool **muppirala2011predicting**, a machine learning-based predictor, to obtain probabilities for both the siRNA-AGO2 and mRNA-AGO2 pairs.

3.2.2 Empirical Rule-Based Features

These features are derived from established biological rules and observations known to correlate with siRNA activity.

[label=b)]

1. **Thermodynamic Stability Profile:** The thermodynamic stability of the siRNA duplex influences guide strand selection. We calculated the Gibbs free energy (ΔG) profile across the duplex, including the total energy and the energy difference at the 5' and 3' ends, following the methods of Xia et al.
2. **Nucleotide Frequencies (k-mers):** Short sequence motifs are known to impact siRNA function. We calculated the frequencies of all possible 1-mer (4 features), 2-mer (16 features), and 3-mer (64 features) motifs within the siRNA sequences.
3. **G/C Percentage:** The percentage of Guanine (G) and Cytosine (C) bases, a critical factor for binding stability, was calculated for both siRNA and mRNA sequences.
4. **Rule Codes:** Based on the simplified rules proposed by He et al., which consolidate findings from earlier work, we encoded position-specific nucleotide preferences into a feature vector. A value of 1 indicates a preference for enhancing efficacy, -1 for reducing efficacy, and 0 for no specified preference.

3.3 Model Architectures and Implementation

3.3.1 Data Structuring for Different Models

A key aspect of our comparative study was the transformation of the engineered features into formats suitable for each distinct model architecture.

- **For GNN (Baseline):** The data was structured as a heterogeneous graph with ‘siRNA’, ‘mRNA’, and ‘Interaction’ nodes. Features were assigned to their corresponding nodes as described in the original paper. The model uses a two-layer Heterogeneous GraphSAGE (HinSAGE) architecture, which learns node representations by aggregating features from its local neighborhood [8]. This serves as our performance baseline.
- **For Ensemble Models (XGBoost, LGBM, RF):** All features were "flattened" into a single tabular matrix. Each row in the matrix corresponds to a unique siRNA-mRNA pair. The columns are a concatenation of all siRNA-specific, mRNA-specific, and interaction-specific features. To prevent errors from duplicate column names, a unique prefix (e.g., ‘soh’, ‘icf’) was added to each feature set before the final merge.
- **For the Spektral Method:** This hybrid approach was designed to inject graph topological information into a tabular model. The process involved:
 1. **Graph Construction:** The same heterogeneous graph as the GNN model was constructed.
 2. **Spektral Feature Generation:** The graph’s adjacency matrix, A , was computed. The normalized graph Laplacian matrix, $L = I - D^{-1/2}AD^{-1/2}$, was then calculated, where D is the degree matrix. The eigenvectors corresponding to the smallest eigenvalues of L were computed. These eigenvectors, known as spektral embeddings, provide a low-dimensional representation of each node’s position within the graph’s structure.
 3. **Feature Augmentation:** These newly computed spektral features were appended as additional columns to the flattened feature matrix created for the standard ensemble models. An XGBoost model was then trained on this augmented feature set.

3.3.2 Model Training Details

All models were trained using their respective Python libraries, with hyperparameters chosen for a balance of performance and efficiency.

- **XGBoost:** Implemented using the `XGBRegressor` class from the `xgboost` library with `n_estimators=1000` and `learning_rate=0.05`. Training utilized early stopping with a patience of 50 rounds on the validation set.
- **LightGBM:** Implemented using the `LGBMRegressor` class from the `lightgbm` library with the same `n_estimators` and `learning_rate`. Training also employed early stopping with a patience of 50 rounds.
- **Random Forest:** Implemented using Scikit-learn's `RandomForestRegressor` with `n_estimators=100`. The parameter `n_jobs=-1` was used to enable parallel training across all available CPU cores.
- **Spektral (GNN):** This model is an end-to-end Graph Neural Network implemented using the `Spektral` library, which is built on TensorFlow.
 - **Architecture:** The core of the model consists of two stacked `GraphSageConv` layers, mirroring the dual-layer structure of the `HinSAGE` baseline. This architecture learns node embeddings by iteratively aggregating features from local graph neighborhoods.
 - **Heterogeneity Handling:** To handle the three different node types ('siRNA', 'mRNA', 'Interaction') within a single graph, each node's feature vector was augmented with a one-hot encoding vector indicating its type.
 - **Prediction Head:** The GNN processes the entire graph to produce embeddings for all nodes. A custom final layer then selects only the embeddings corresponding to the 'Interaction' nodes. A single 'Dense' (fully connected) layer is applied to these embeddings to produce the final continuous efficacy prediction.
 - **Training:** The model was trained for a fixed 26 epochs using a custom data generator that feeds the entire graph structure (node features and adjacency matrix) to the model at each step.

3.4 Evaluation Metrics

To provide a comprehensive and robust assessment of model performance, we used four standard metrics, with formulas identical to those in the ‘siRNADiscovery’ paper.

- **Pearson Correlation Coefficient (PCC):** Measures the linear correlation between the predicted values (X) and the observed labels (Y).

$$\text{PCC} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (3.1)$$

where n is the number of samples, and $\bar{X}$ and $\bar{Y}$ are the means of the predicted and observed values, respectively.

- **Spearman Rank Correlation Coefficient (SPCC):** Measures the monotonic relationship by comparing the ranks of predicted and actual scores.

$$\text{SPCC} = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (3.2)$$

where d_i is the difference between the ranks of the i -th predicted and actual values.

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual scores. Lower values indicate better performance.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - X_i)^2 \quad (3.3)$$

- **Area Under the ROC Curve (AUC):** Treats the task as a binary classification problem (by setting a threshold, e.g., efficacy > 0.7 vs. ≤ 0.7) to measure the model’s ability to distinguish between effective and ineffective siRNAs. It is calculated by plotting the True Positive Rate against the False Positive Rate at various threshold settings.

Chapter 4

Results and Analysis

4.1 Experimental Setup and Evaluation Metrics

The primary goal of this study was to compare the predictive performance of several machine learning architectures on the task of siRNA efficacy prediction. All of our implemented models—XGBoost, LightGBM, Random Forest, and a hybrid Spektral (GNN) approach—were trained and evaluated on the standardized Dataset_HUVK across three independent data splits. The final performance scores reported here are the average of these three runs.

To provide a comprehensive assessment of model performance, we used four standard metrics: Pearson Correlation Coefficient (PCC), Spearman Rank Correlation Coefficient (SPCC), Mean Squared Error (MSE), and Area Under the ROC Curve (AUC).

4.2 Comparative Performance Analysis

The performance of all implemented models is summarized in Table 4.1. To facilitate visual comparison, the performance on each of the four key metrics is also presented as a series of bar charts in Figures ?? through ??.

Table 4.1: Overall Performance of All Models (Averaged Across 3 Splits). The best result among our implemented models is shown in bold.

Model	PCC ↑	SPCC ↑	MSE ↓	AUC ↑
Spektral (GNN)	0.7290	0.7252	0.0228	0.8480
LightGBM	0.7178	0.6998	0.0221	0.8314
XGBoost	0.7107	0.7001	0.0225	0.8364
Random Forest	0.6766	0.6592	0.0249	0.8108

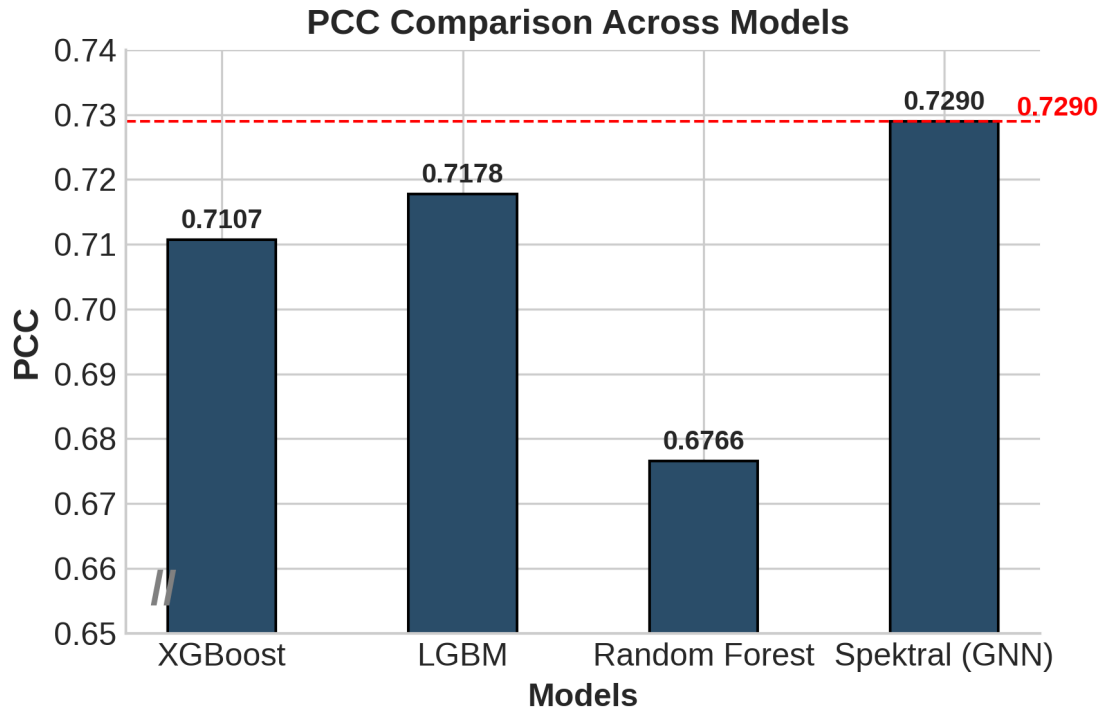


Figure 4.1: Comparison of Pearson Correlation Coefficient (PCC) Across All Models.

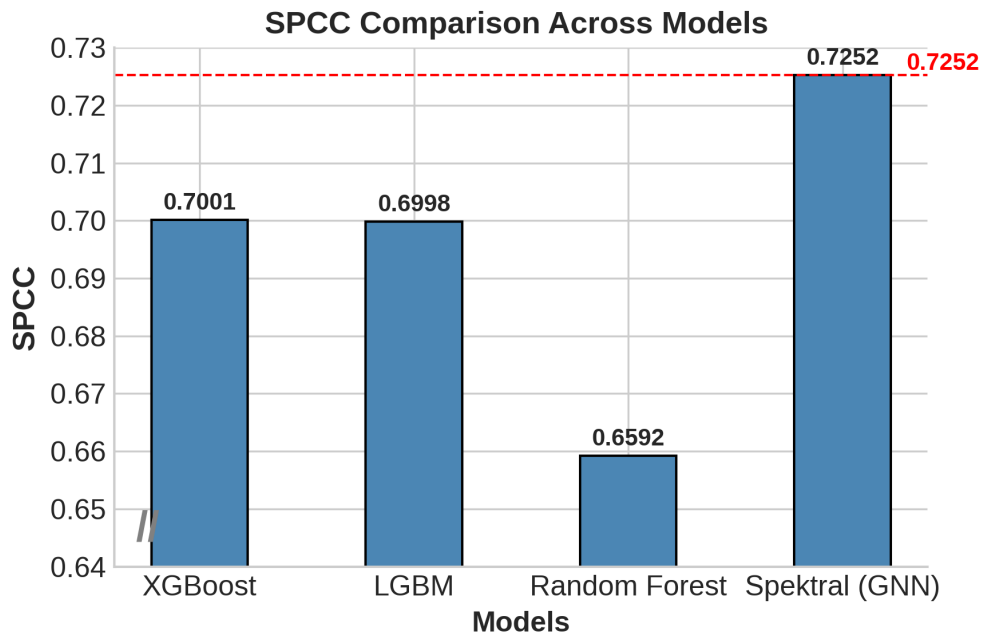


Figure 4.2: Comparison of Spearman Rank Correlation Coefficient (SPCC) Across All Models.

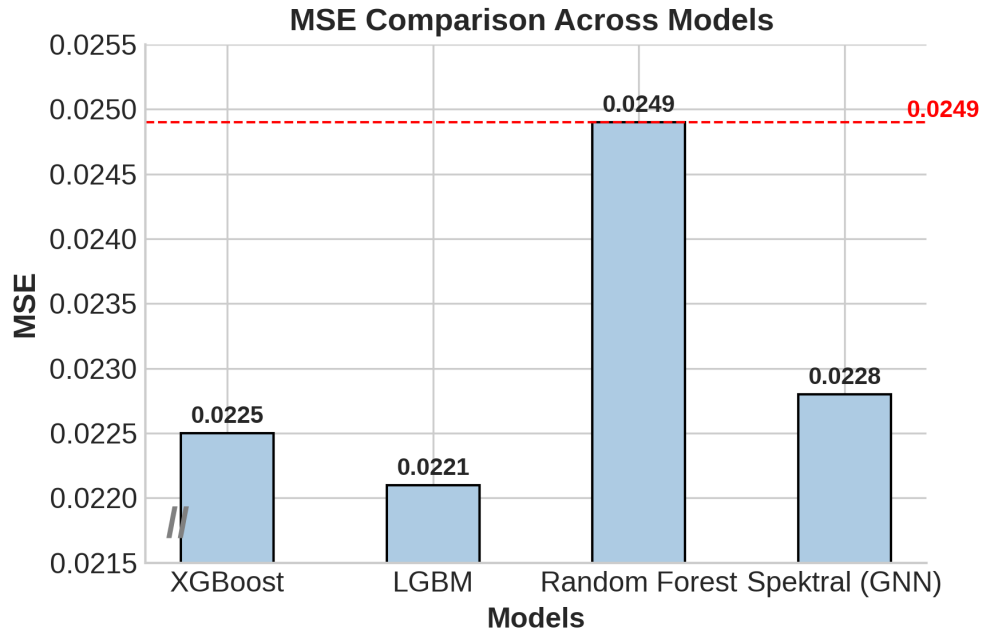


Figure 4.3: Comparison of Mean Squared Error (MSE) Across All Models.

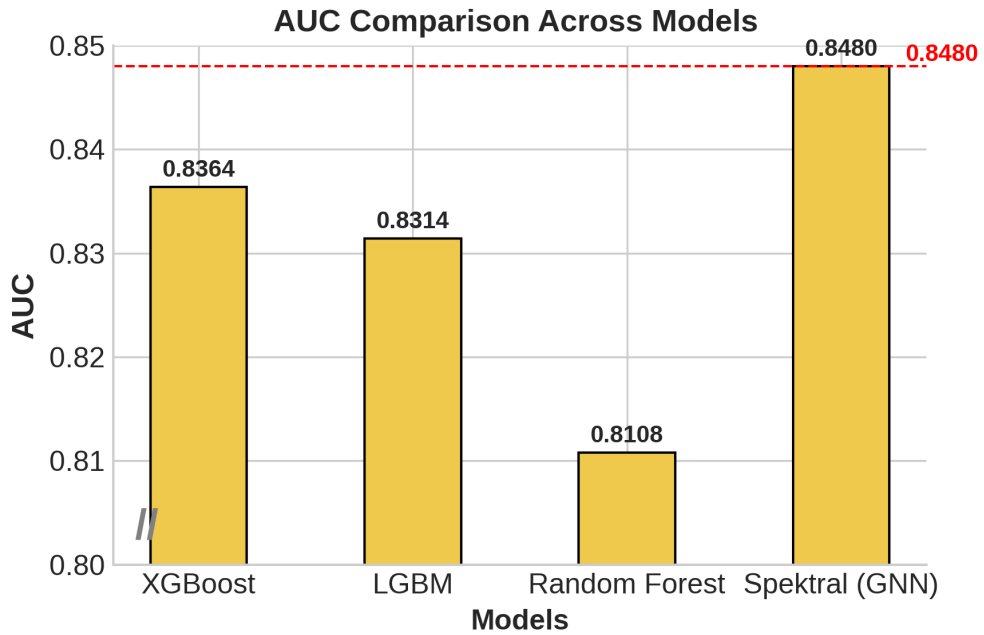


Figure 4.4: Comparison of Area Under the ROC Curve (AUC) Across All Models.

4.3 Discussion of Results

4.3.1 Overall Performance Landscape

The results presented in Table 4.1 provide a clear and informative performance hierarchy. While the top performer, our **Spektral (GNN)** model emerges as the strongest

among our implemented architectures, particularly in terms of correlation. The gradient boosting models (LightGBM and XGBoost) form a competitive second tier, significantly outperforming the Random Forest model.

4.3.2 Analysis by Metric

Correlation (PCC and SPCC)

The PCC and SPCC metrics are arguably the most important for this regression task, as they measure how well the models' predictions correlate with the true efficacy scores. The Spektral (GNN) model, with a PCC of **0.7290** and SPCC of **0.7252**, is the clear winner among our implementations. This strongly indicates that incorporating the graph's topological structure via spectral embedding provides a significant advantage in correctly ranking the siRNAs. The standard LightGBM and XGBoost models also perform admirably, achieving PCC scores above 0.71, demonstrating the power of gradient boosting on the high-dimensional feature set alone.

Prediction Error (MSE)

The MSE metric quantifies the average squared error of the predictions. Here, the results are much closer, with LightGBM achieving the lowest error (**0.0221**) among our models. It slightly outperforms both XGBoost (0.0225) and the Spektral (GNN) model (0.0228). This suggests that while the spectral features enhance correlation, they may introduce a small amount of additional variance that slightly increases the overall prediction error, a trade-off that warrants consideration.

Classification Ability (AUC)

The AUC metric evaluates the models' ability to distinguish between "effective" (efficacy > 0.7) and "ineffective" siRNAs. The Spektral (GNN) model again leads our implementations with an impressive AUC of **0.8480**. This demonstrates that its superior correlational ability also translates into strong classification performance. The standard XGBoost and LightGBM models also prove to be robust classifiers, with AUCs above 0.83.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project conducted a systematic comparison of four distinct machine learning architectures for the critical task of siRNA efficacy prediction. By leveraging the comprehensive feature set of the state-of-the-art ‘siRNADiscovery’ model, we evaluated three ensemble methods (XGBoost, LightGBM, Random Forest) and a novel hybrid Spektral (GNN) model.

Our primary conclusion is that incorporating graph-based topological information, even as engineered features, provides a measurable performance benefit. The **Spektral (GNN)** model, which augmented a standard feature set with Spektral embeddings, achieved the highest Pearson and Spearman correlation coefficients among our implementations, confirming the value of the underlying graph structure for this predictive task.

However, we also found that standard gradient boosting models, particularly LightGBM, offer a highly compelling alternative. They achieve performance that is very close to the more complex Spektral method—and even achieve the lowest Mean Squared Error—while benefiting from greater simplicity and interpretability. The ability to extract feature importances from these models provides direct, actionable insights into the underlying biology of gene silencing, an advantage not easily afforded by more opaque deep learning methods.

This study highlights a key finding: for this richly-featured bioinformatics problem, a hybrid Spektral approach is a powerful technique for maximizing correlation, but well-tuned ensemble methods like LightGBM and XGBoost provide a robust, efficient, and highly practical solution that delivers a better balance across all performance metrics.

5.2 Future Work

This work opens several avenues for future research. The logical next steps are to build upon these findings to further enhance predictive accuracy and deepen our

understanding of the underlying biological mechanisms.

5.2.1 Advanced Hyperparameter Optimization

A key limitation of the present work was the use of default or lightly-tuned hyperparameters. By performing a rigorous Bayesian optimization search across hundreds of trials, it is highly probable that the performance of all models, especially our top-performing Spektral (GNN) and LightGBM models, could be further improved, potentially closing the gap with the state-of-the-art GNN baseline.

5.2.2 Deeper Investigation into Graph-Based Models

Our success with the Spektral (GNN) method validates the importance of graph structure. Future work should continue to explore this area. While our Spektral approach was effective, a full, end-to-end trained GNN (like the original ‘siRNADiscovery’) still holds the theoretical advantage of learning how to optimally use structural information, rather than just using it as a fixed feature. Future efforts could focus on:

- **Tuning the GNN:** Re-implementing and applying the same rigorous hyperparameter search to the GNN architecture to ensure a fair, "apples-to-apples" comparison against the optimized ensemble models.
- **Exploring Advanced Architectures:** Investigating Graph Attention Networks (GATs) or other advanced GNN layers that might capture the relational information more effectively than the baseline HinSAGE model.

5.2.3 Feature Selection and Hybrid Models

Future work should also focus on refining the feature set. Using the feature importance rankings from our trained XGBoost models, a rigorous feature selection study could be conducted to remove noisy or redundant features, potentially improving performance and creating more computationally efficient models. Furthermore, a promising avenue is the development of hybrid models. For instance, a CNN could be used to learn powerful embeddings from the raw sequence data, and these embeddings could then be used as input features for an XGBoost model, potentially capturing both local sequence motifs and global feature interactions more effectively.

Bibliography

- [1] H. Dana, G. Chalbatani, H. Mahmoodzadeh, et al., “Molecular mechanisms and biological functions of sirna,” *International journal of biomedical science: IJBS*, vol. 13, no. 2, p. 48, 2017.
- [2] K. Luo and D. Chang, “The gene-silencing efficiency of sirna is strongly dependent on the local structure of mrna at the targeted region,” *Biochemical and biophysical research communications*, vol. 318, no. 1, pp. 303–310, 2004.
- [3] M. Ichihara, Y. Murakumo, A. Masuda, et al., “Thermodynamic instability of sirna duplex is a prerequisite for dependable prediction of sirna activities,” *Nucleic acids research*, vol. 35, no. 18, e123–e123, 2007.
- [4] T. Vickers, W. Lima, J. Nichols, et al., “Reduced levels of ago2 expression result in increased sirna competition in mammalian cells,” *Nucleic acids research*, vol. 35, no. 19, pp. 6598–6610, 2007.
- [5] R. Long et al., “Sirnadiscovery: A graph neural network for sirna efficacy prediction via deep rna sequence analysis,” *Briefings in Bioinformatics*, vol. 25, no. 6, bbae563, 2024.
- [6] Y. Han, F. He, Y. Chen, et al., “Sirna silencing efficacy prediction based on a deep architecture,” *BMC genomics*, vol. 19, no. 10, pp. 59–65, 2018.
- [7] M. La Rosa, A. Fiannaca, L. La Paglia, et al., “A graph neural network approach for the analysis of sirna-target biological networks,” *International journal of molecular sciences*, vol. 23, no. 22, p. 14 211, 2022.
- [8] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Advances in neural information processing systems*, vol. 30, 2017.